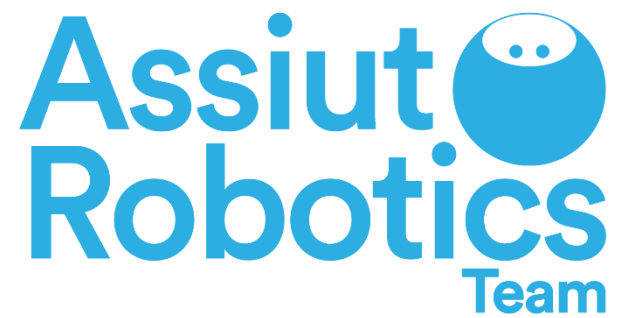


Arduino basics



Week #01

Main Objectives:

- Take an overview of the Arduino board, its components, its advantages, and how it is useful
- Know the needed items to start learning Arduino (Hardware, Software)
- Know how to learn Arduino via tinkercad
- Hands on your first Arduino project
- Good understanding of basic Arduino functions

Study material:

- [Simply Arduino \(Book\)](#) -> from page 7 to page 69
- [Arduino course](#) -> 01 ~ 06

1st_Task:

Develop a code to control an LED using two push buttons, one button should turn on the LED when it has been pressed, and the other turn off the LED

The code must be clean and comments must clarify every line

2nd_Task:

Develop a code to simulate a traffic light,

You have to use 3 LEDs (Green, Yellow, Blue), and each LED must be declared as a variable in your code, the performance of your code must follow that flow:

- 1- Green LED starts for 3 sec
- 2- Turn off the green LED and start the yellow LED for 3 sec
- 3- Turn off the yellow LED and start the red LED for 3 sec

4- Turn off the red LED and the loop start over again

BONUS: connect a push button to your circuit, as the traffic light will make one loop every time the button is pressed

3rd_Task:

Develop a code to simulate a traffic light,

You have one RGB LED, you should declare every color as a variable, also you have to implement a function with 3 parameters to light the RGB LED, the performance of your code must follow that flow:

- 1- Green LED starts for 3 sec
- 2- Yellow LED starts for 3 sec
- 3- Red LED starts for 3 sec

Turn off the red LED and the loop start over again